Jan Škvrna

Generative 3D Reconstruction & Weakly Supervised 3D Perception
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jskvrna.github.io | GitHub | Google Scholar | LinkedIn
Available for research internships in 2027 with flexible timing and location.

I am a CTU Prague PhD student with **four first-author papers** (ECCV, ICCV, IJCV, WACV 2027 accepted). My teams placed **first in two S23DR challenges** at the CVPR Urban Scene Modeling Workshop.

Education & Research

PhD in Informatics Feb 2024 – Expected Feb 2028
CTU Prague, Visual Recognition Group, advisor Dr. Lukáš Neumann
MSc in Cybernetics and Robotics, with Honors 2021 – 2024
CTU Prague, Dean's Award for Master's Thesis

Publications & Research Contributions

WireDiT: A Vertex-Slot Diffusion Transformer for 3D Wireframe Reconstruction
***J. Škvrna**, M. Purkrábek, L. Neumann.* **WACV 2027, accepted**.

- Designed and implemented a flow-matching transformer with 64 vertex slots conditioned on point-anchored scene tokens to predict 3D vertices and edges from sparse street-level SfM points, depth, and semantics.
- Achieved **23% lower average corner offset** on Building3D Tallinn (0.17 m vs. 0.22 m, EdgeDiff+GREO). On HoHo, a two-stage pipeline with 16-sample consensus took **5.5 s/scene on one A100**, including preprocessing.

MonoSOWA++: Universal Autolabeling Pipeline for Monocular 3D Object Detection
***J. Škvrna**, L. Neumann.* **IJCV 2026**. Paper | Code

- Extended monocular 3D autolabeling to pedestrians and cyclists using predicted depth, 2D detections and confidence scores. End-to-end pseudo-label generation takes **1.9 s/frame on KITTI-360**.

MonoSOWA: Scalable Monocular 3D Object Detector Without Human Annotations
***J. Škvrna**, L. Neumann.* **ICCV 2025**. Paper | Code

- Separated object/camera motion and normalized focal length for monocular autolabeling. After pretraining on KITTI-360 pseudo-labels and fine-tuning on KITTI, MonoDETR outperformed the same architecture trained from scratch with all KITTI labels, while using **75% fewer KITTI labels** (Car, moderate 3D AP, IoU 0.7).

TCC-Det: Temporarily Consistent Cues for Weakly-Supervised 3D Detection
***J. Škvrna**, L. Neumann.* **ECCV 2024**. Paper | Code

- Combined temporal consistency and robust 3D template fitting for LiDAR detection on KITTI and the Waymo Open Dataset. Without human KITTI 3D box annotations, the method reached **about 90% of the fully supervised baseline's 3D AP** on KITTI (Car, moderate, IoU 0.7).

Awards

1st place, S23DR 3D Reconstruction Challenge (US$5,000 team prize) 2026
CVPR Urban Scene Modeling Workshop, WireDiT solution, team of three. Report | Project
1st place, S23DR 3D Reconstruction Challenge (US$10,000 team prize) 2025
CVPR Urban Scene Modeling Workshop, two-stage PointNet pipeline, team of two. Report | Project

Experience & Service

Junior Embedded Software Developer, Poll s.r.o. 2020 – 2023
Real-time C/C++ firmware for ARM microcontrollers in HVAC and high-voltage DC/DC converters.
Lab Instructor, Deep Learning Essentials, CTU Prague 2025 – Present
Reviewer: ICCV 2025, BMVC 2026, AAAI 2027.

Technical Skills & Current Research

Research: Diffusion and flow matching, multi-view geometry, 3D reconstruction and object detection. Current research focuses on weakly supervised semantic 3D occupancy using temporal consistency.
Engineering: Python, PyTorch, multi-GPU training, Slurm experiment automation, self-hosted apps, dashboards and agents, C/C++, Open3D, PyTorch3D, OpenCV.
Invited talks: CMP Colloquium (2026), CVWW (2025). **Summer schools:** EEML (2026), ICVSS (2025).